

Pranathi Deepak

Los Angeles, CA (Open to Relocation)

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Summary

Machine Learning Engineer with 4+ years of experience designing, training, and deploying production ML and AI systems at scale. Proven track record building end-to-end pipelines from feature engineering and model training to real-time inference services and MLOps automation using Python, PyTorch, and AWS. Hands-on experience with Generative AI, Agentic AI systems, LLMs, and RAG architectures including autonomous multi-step AI agents built with LangChain Agents and LlamaIndex. Strong MLOps foundation: CI/CD, experiment tracking, drift detection, and scalable deployment via Docker and Kubernetes. Delivered measurable business impact across financial services (KPMG) and applied AI environments.

Technical Skills

Languages: Python, SQL, Java, R

ML & AI: XGBoost, Random Forest, Gradient Boosting, SVM, Clustering, Scikit-learn, Hyperparameter Tuning, SHAP, ROC-AUC

Generative & Agentic AI: Generative AI, Agentic AI, AI Agents, LLMs, RAG (Retrieval-Augmented Generation), LangChain, LangChain Agents, LlamaIndex, FAISS, Pinecone, Prompt Engineering, Tool Use, Multi-Step Reasoning

Deep Learning & NLP: PyTorch, TensorFlow, CNNs, RNNs, LSTMs, Transformers, Embeddings, Text Classification, Semantic Search

MLOps & Cloud: AWS (S3, EC2, SageMaker, Lambda, Redshift), Azure ML, GCP Vertex AI, Docker, Kubernetes, Airflow, MLflow, CI/CD, Model Versioning, Drift Detection

Data & Streaming: Pandas, NumPy, Spark, Spark Streaming, Kafka, Databricks, ETL/ELT Pipelines, Feature Stores, Data Validation

APIs & Deployment: FastAPI, Flask, REST APIs, Batch Inference, Real-Time Inference, Model Serving

Work Experience

Machine Learning Engineer

Aug 2024 – Dec 2025

California State University

Carson, CA

- Architected and deployed RAG (Retrieval-Augmented Generation) pipelines using LangChain, LlamaIndex, and FAISS, enabling context-aware LLM responses over large-scale structured and unstructured datasets, reducing information retrieval time by 60%.
- Designed end-to-end ML pipelines using Python, Pandas, SQL, and AWS S3, automating data ingestion, validation, and feature transformation, reducing manual data preparation effort by 60%.
- Built embedding-based semantic search systems using transformer models, improving retrieval relevance and enabling scalable knowledge discovery across large datasets.
- Built and optimized supervised learning models (XGBoost, Random Forest, Logistic Regression) with cross-validation and Bayesian hyperparameter tuning, improving predictive accuracy by 18% over baseline models.
- Deployed scalable REST inference APIs using FastAPI and Docker, supporting batch and near-real-time prediction workflows for internal stakeholders.
- Implemented full MLOps lifecycle with MLflow experiment tracking, model versioning, and performance comparison, enabling reproducible model governance across 10+ iterations.
- Established automated monitoring pipelines with metric logging and statistical drift detection, reducing undetected model performance drops by 40%.
- Automated batch data pipelines with Python and SQL, ensuring reliable nightly ingestion and preprocessing for downstream model training and inference workflows.

Math Tutor

Oct 2023 – Jul 2024

Davis Middle School

Carson, CA

- Delivered algebra and statistics instruction to 15+ students, contributing to a 20% improvement in average exam scores through structured, data-driven lesson planning.
- Designed problem-solving exercises focused on equations and probability, strengthening student's analytical reasoning — directly translating to skills applied in ML model evaluation and statistical analysis.
- Tracked and analyzed student performance data to identify learning trends and adjust instructional approach, applying the same data-based feedback loops used in ML monitoring workflows.

Machine Learning Engineer

KPMG Global Services

Oct 2020 – Aug 2023

Bengaluru, India

- Built distributed ML pipelines using Python, PySpark, and Databricks to process millions of transactional records daily for predictive analytics and financial risk applications, reducing pipeline runtime by 35%.
- Designed and maintained a centralized feature store on AWS Redshift, ensuring training-serving consistency and eliminating feature drift across 5+ production models.
- Developed and tuned supervised and unsupervised models (XGBoost, SVM, K-Means, PCA) using Bayesian hyperparameter optimization, achieving 15–20% accuracy improvements on client risk-scoring benchmarks.
- Deployed scalable batch and near-real-time inference pipelines on AWS with containerized Docker workflows, supporting high-throughput production ML applications serving enterprise clients.
- Automated CI/CD model deployment pipelines using Git and Docker, with SHAP-based explainability reporting and ROC-AUC / precision-recall evaluation dashboards for stakeholder sign-off.
- Integrated real-time data streaming components using Kafka and Spark Streaming to feed live transactional signals into inference pipelines, reducing model latency by 25%.

Project

Breast Cancer Survival Prediction System

- Engineered a clinical survival analysis platform using Cox Proportional Hazards and Random Survival Forest models (lifelines, scikit-learn, PyTorch), achieving a C-index of 0.81 on held-out patient cohorts.
- Built automated feature engineering pipelines (Pandas, NumPy) with missing-value imputation and normalization for clinical datasets, and deployed a FastAPI inference endpoint on Docker for real-time risk prediction.
- Integrated SHAP explainability dashboards identifying top 5 clinical biomarkers driving patient risk scores, enabling actionable model transparency for medical stakeholders.

Heart Disease Prediction API

- Developed an end-to-end ML classification system using Python and scikit-learn to predict heart disease risk from clinical datasets, achieving 89% AUC on the test set with ensemble model stacking.
- Deployed a Flask REST API with Docker containerization for real-time inference, integrated with SHAP explainability to surface the top predictive features per patient, supporting transparent clinical decision-making.

Education

Master of Science in Computer Science | California State University, Dominguez Hills, Carson, CA

Bachelor of Science in Computer Science | K.S. School of Engineering and Management, Bengaluru, KA